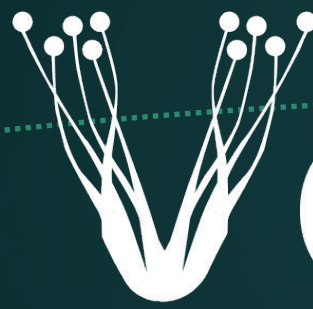




VoxelBox

rs-fMRI Toolkit

Everything you need to get started.

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Sensorimotor Network

Eloquent Networks

FUNCTION

Processes sensory inputs and converts them into motor actions for balance, posture, and voluntary movement

DAMAGE CAUSES

Weakness, paralysis, or loss of motor precision

KEY REGIONS

Primary Motor Cortex (M1)

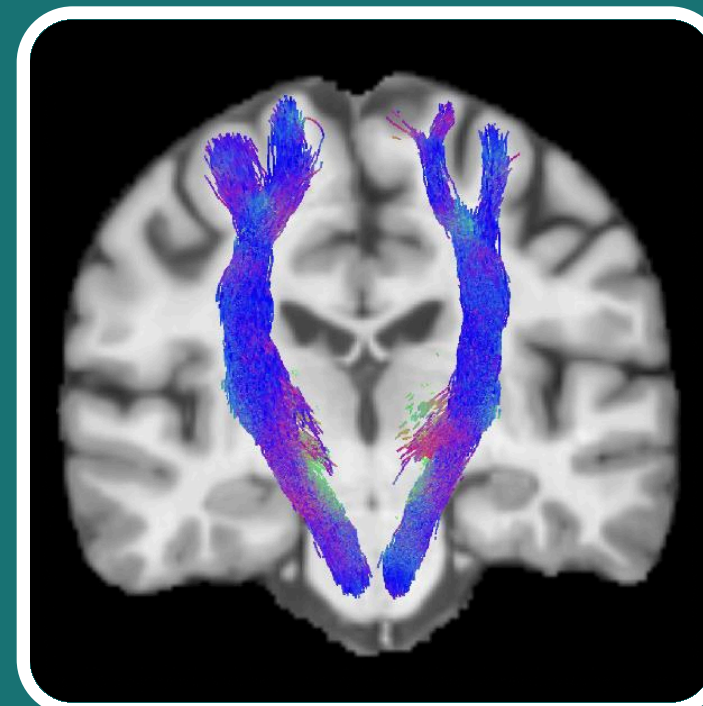
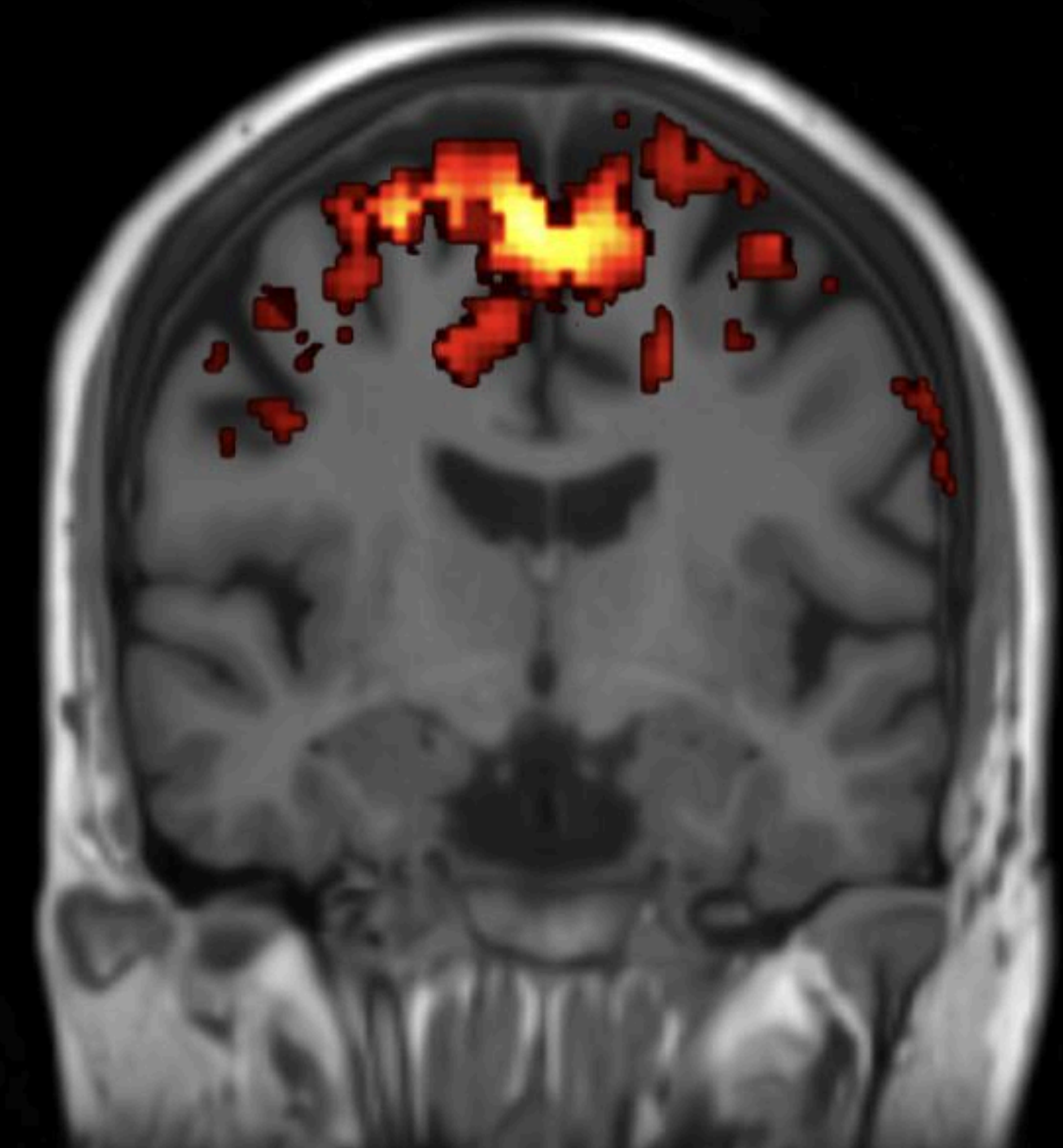
Executes voluntary movements

Primary Somatosensory Cortex (S1)

Processes sensory inputs

Premotor Cortex & SMA

Plans and initiates movements



KEY PATHWAYS

Corticospinal Tract

Carries motor signals from the brain to the spinal cord

Language Network

Eloquent Networks

FUNCTION

Handles speech production, comprehension, and written language

DAMAGE CAUSES

Aphasia, loss of comprehension or speech disabilities

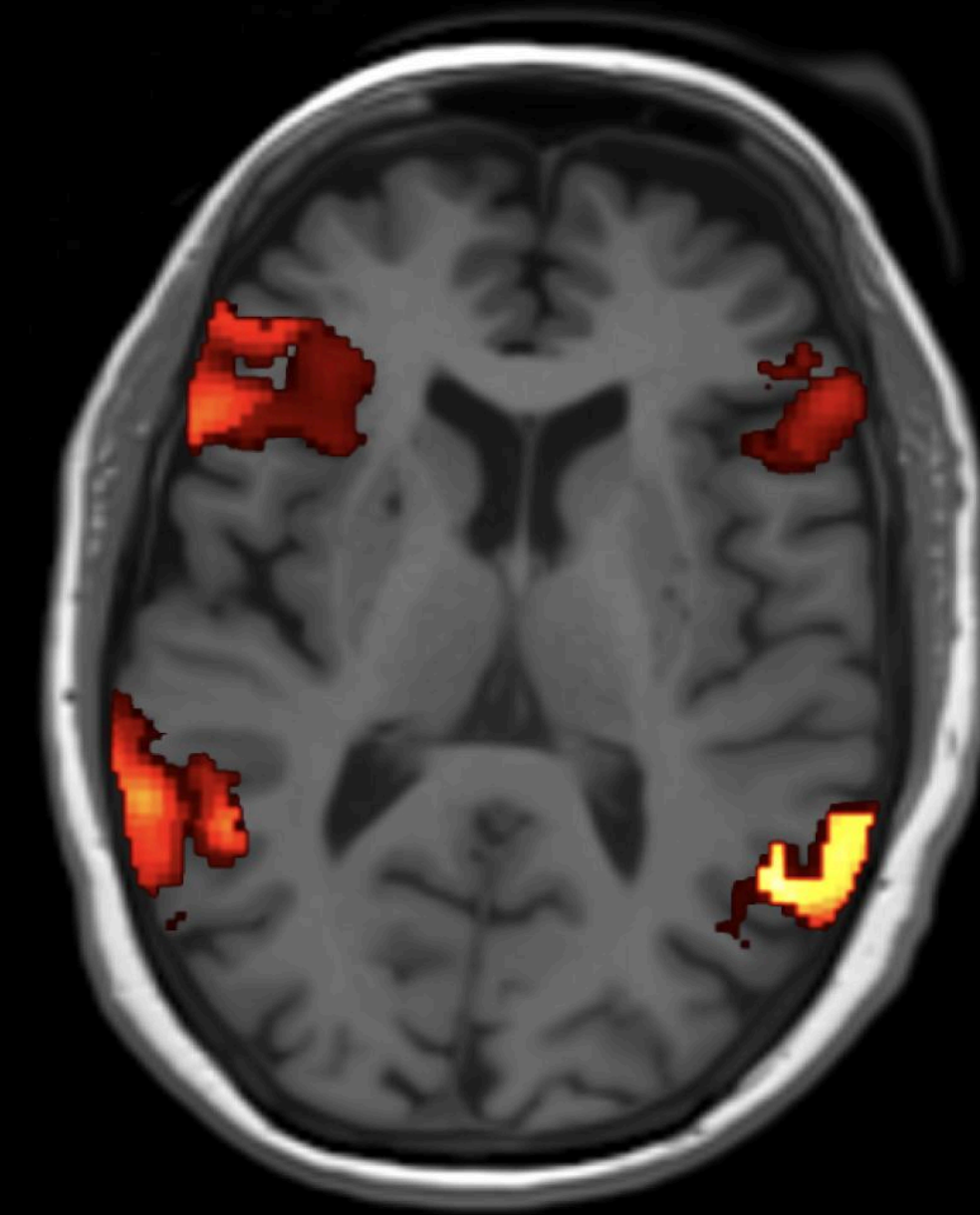
KEY REGIONS

Broca's area

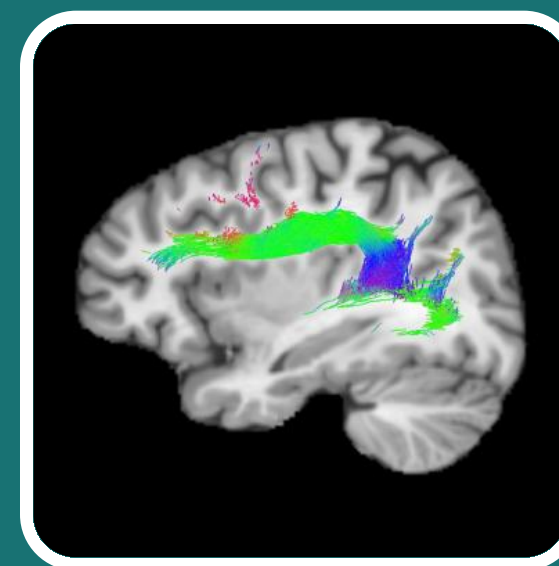
Speech production and grammar

Wernicke's area

Language comprehension

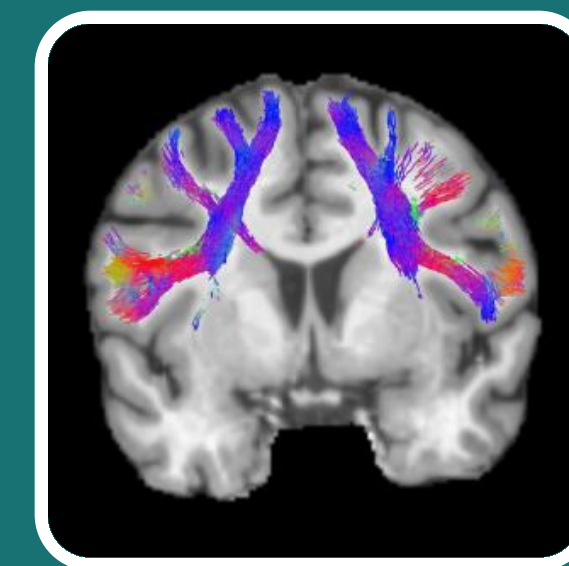


KEY PATHWAYS



Arcuate Fasciculus

Connects Broca's & Wernicke's areas



Frontal Aslant Tracts

Supports speech initiation & verbal fluency

Visual Network

Eloquent Networks

FUNCTION

Processes visual information like object recognition, spatial perception, and movement.

DAMAGE CAUSES

Vision loss, field defects, or difficulty in visual recognition

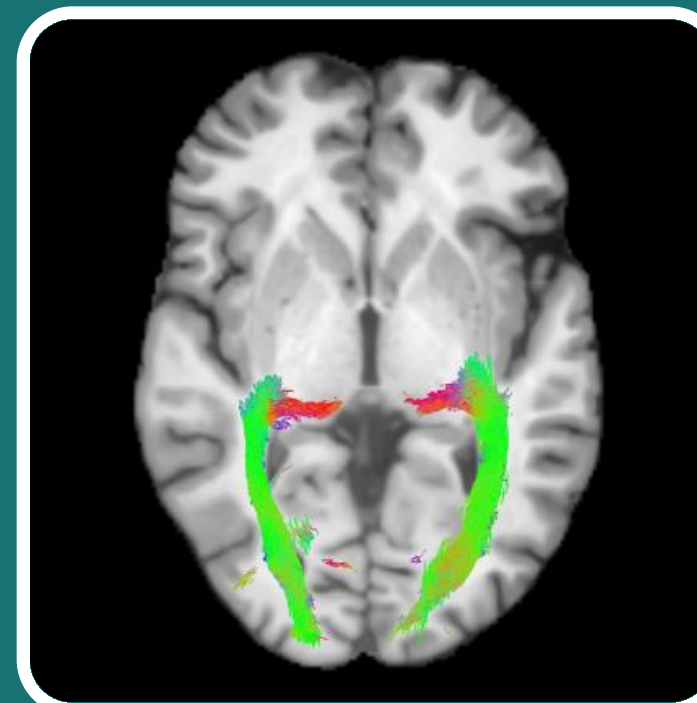
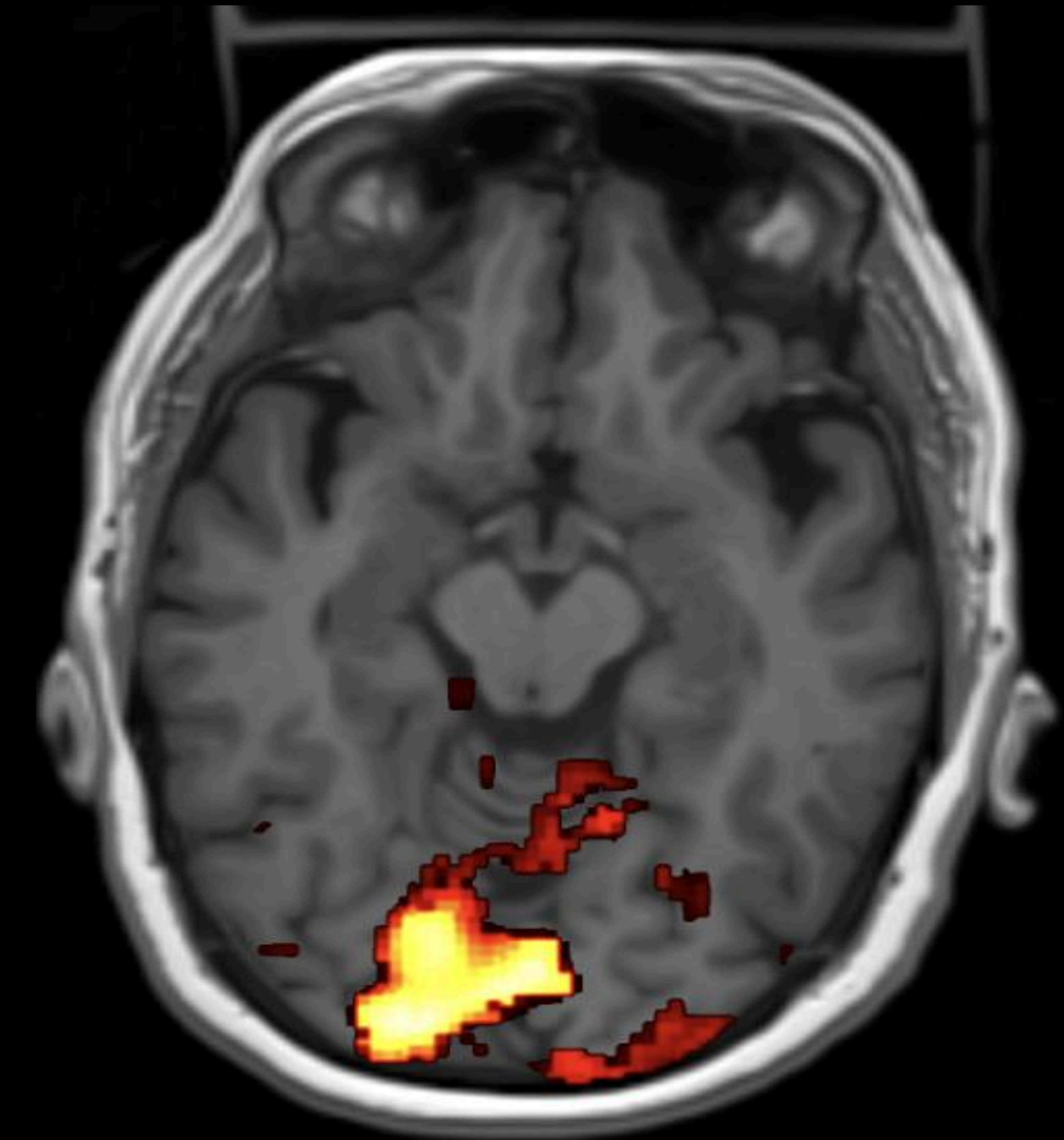
KEY REGIONS

Primary Visual Cortex (V1)

Processes raw visual data

Visual Association Areas

Interprets and integrates visual signals



KEY PATHWAYS

Optic Radiation

Carries visual signals from the thalamus to the visual cortex. Carries visual signals from the thalamus to the visual cortex.

Auditory Network

Eloquent Networks

FUNCTION

Processes sound, speech, and auditory signals for communication and awareness

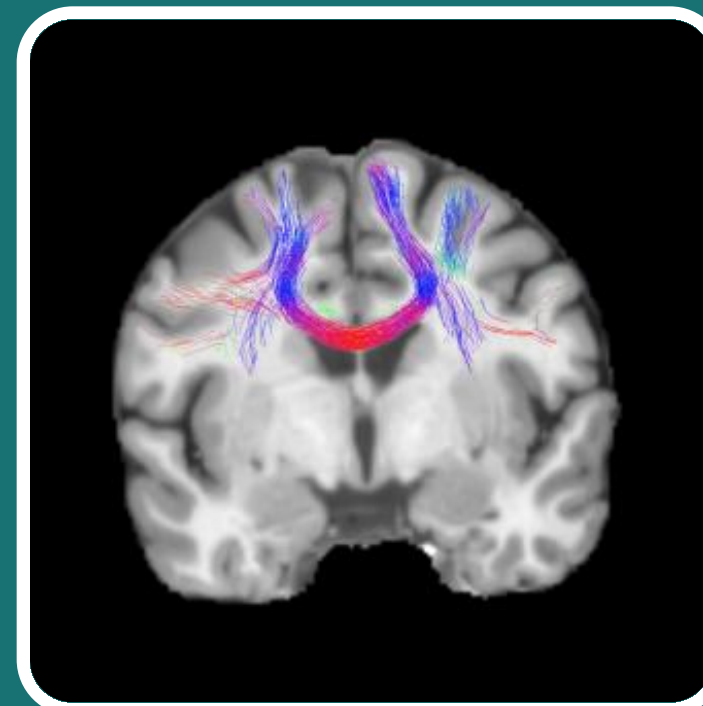
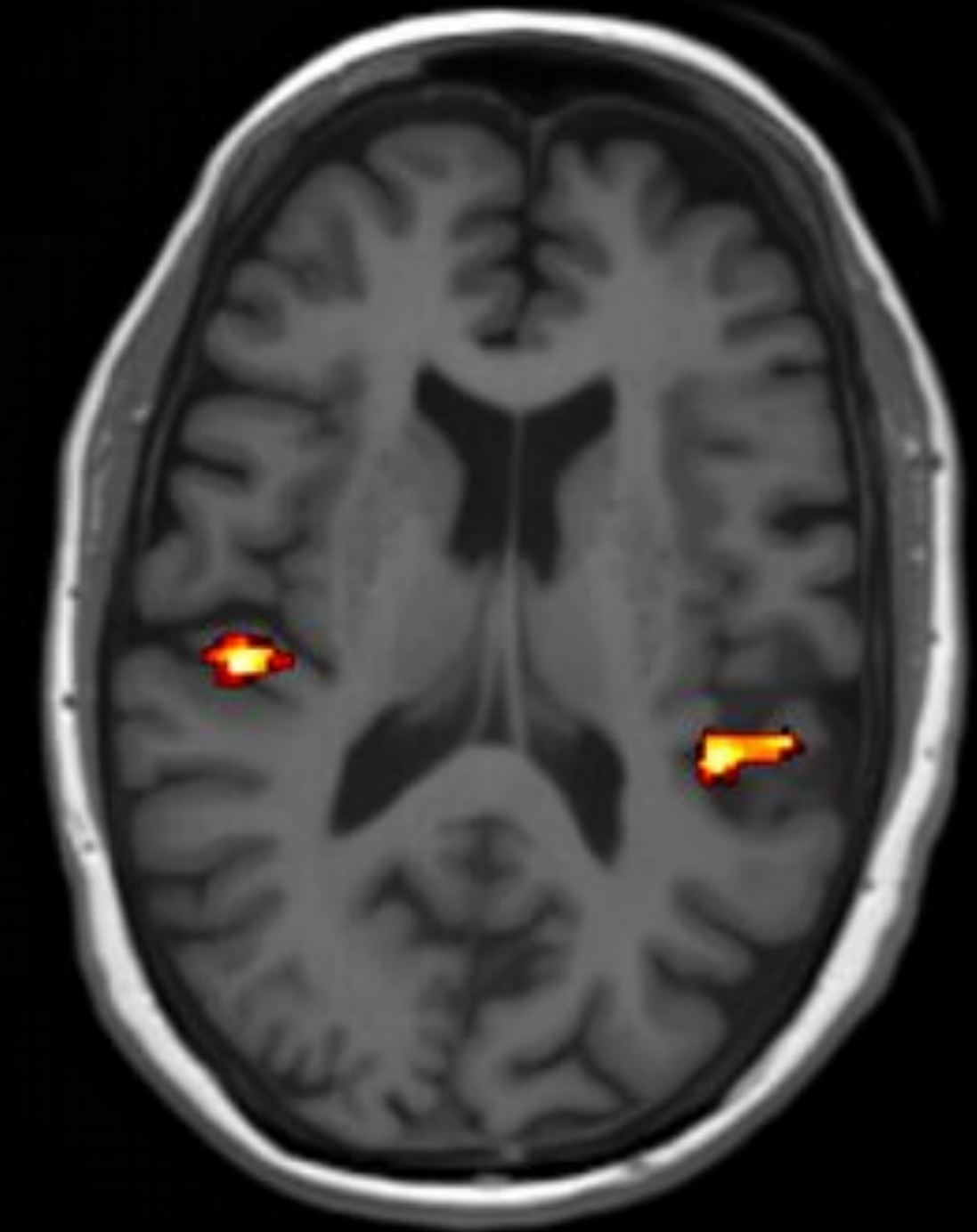
DAMAGE CAUSES

Hearing loss or difficulty in sound localization and comprehension

KEY REGIONS

Primary Auditory Cortex

Processes sound signals



KEY PATHWAYS

Corpus Callosum

Connects the primary auditory cortices bilaterally

Default Mode Network (DMN)

Cognitive Networks

FUNCTION

Supports internal thoughts, daydreaming, self-reflection, and autobiographical memory retrieval

DAMAGE CAUSES

Impaired memory, attention deficits, or self-awareness issues

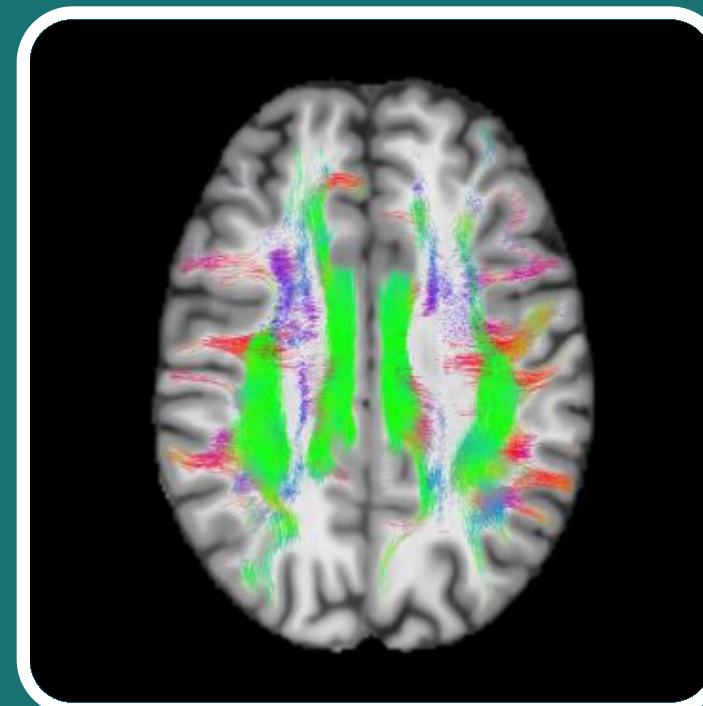
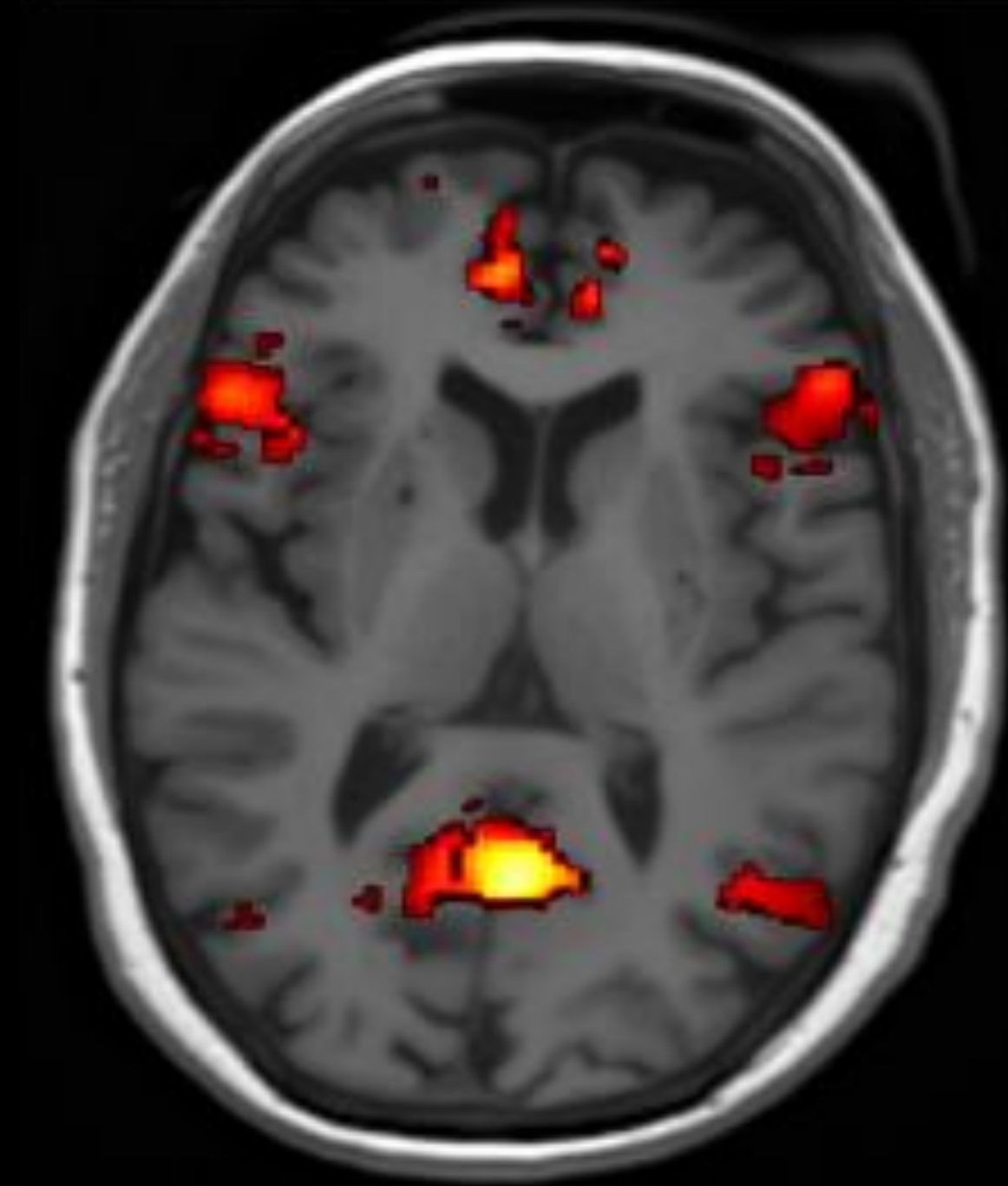
KEY REGIONS

Medial Prefrontal cortex

Posterior Cingulate Cortex

Medial Temporal Lobe

Inferior Parietal Lobule



KEY PATHWAYS

Superior Longitudinal Fasciculus

Central Executive Network (CEN)

Cognitive Networks

FUNCTION

Controls working memory, decision-making, goal-directed tasks and coordination of various cognitive processes, especially those related to higher-level executive functions

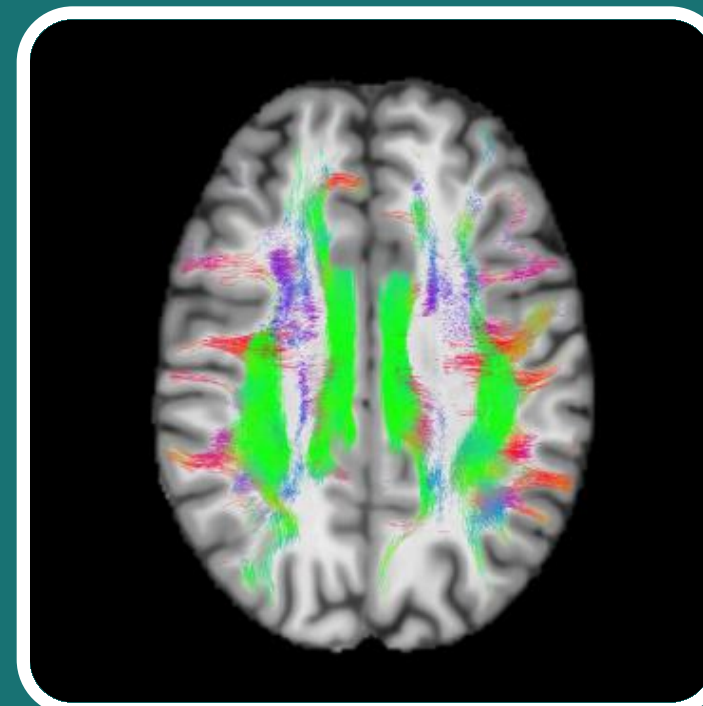
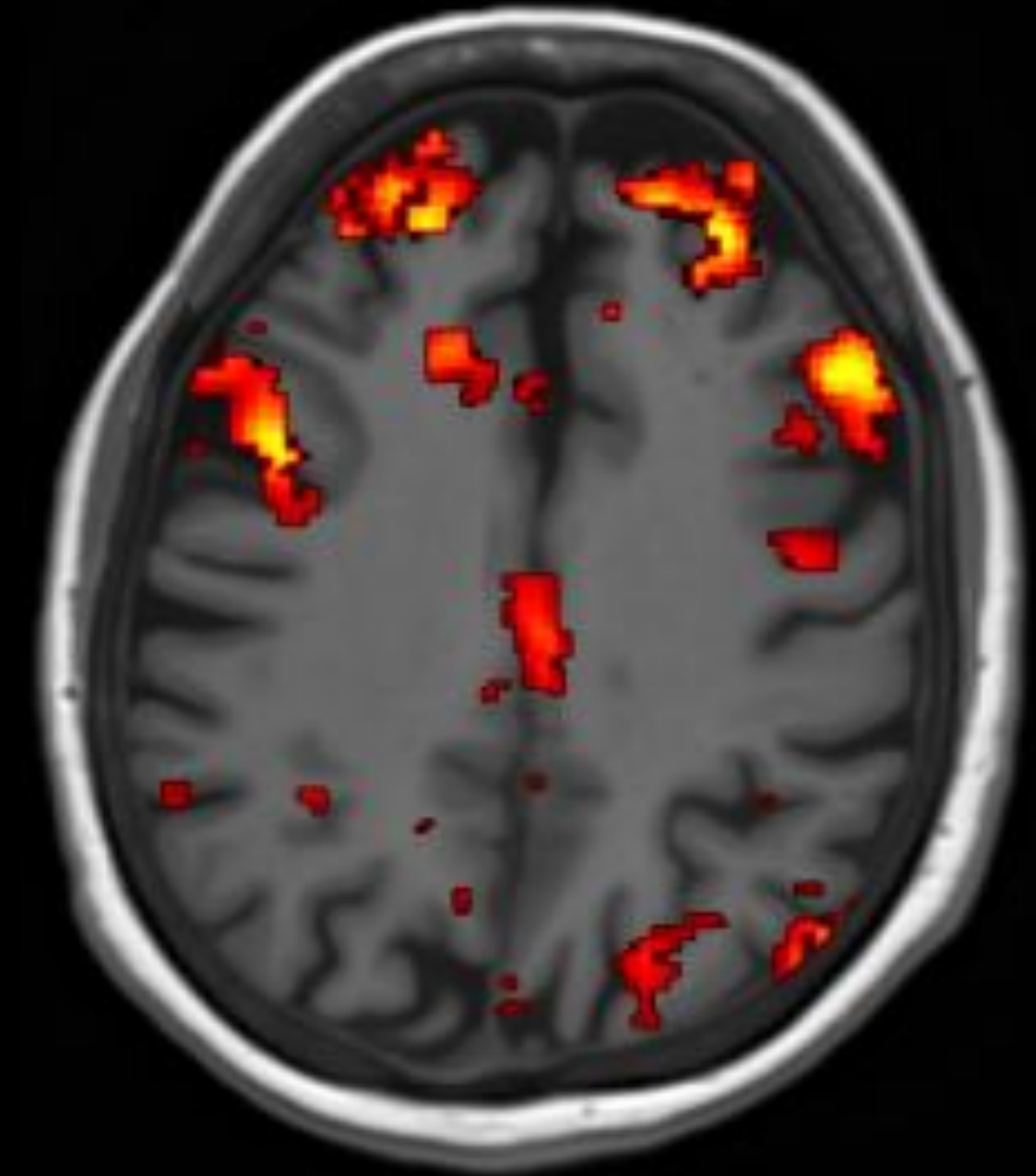
DAMAGE CAUSES

Poor decision-making, reduced working memory, or executive dysfunction.

KEY REGIONS

Dorsolateral Prefrontal Cortex (DLPFC)

Posterior Parietal Cortex



KEY PATHWAYS

Superior Longitudinal Fasciculus

Salience Network

Cognitive Networks

FUNCTION

Detects and orients attention toward salient or behaviourally relevant stimuli in environment

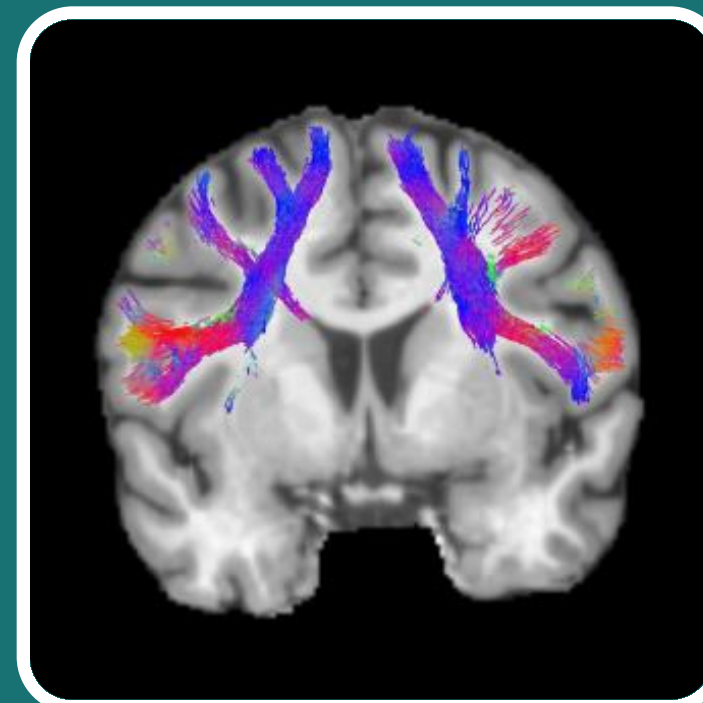
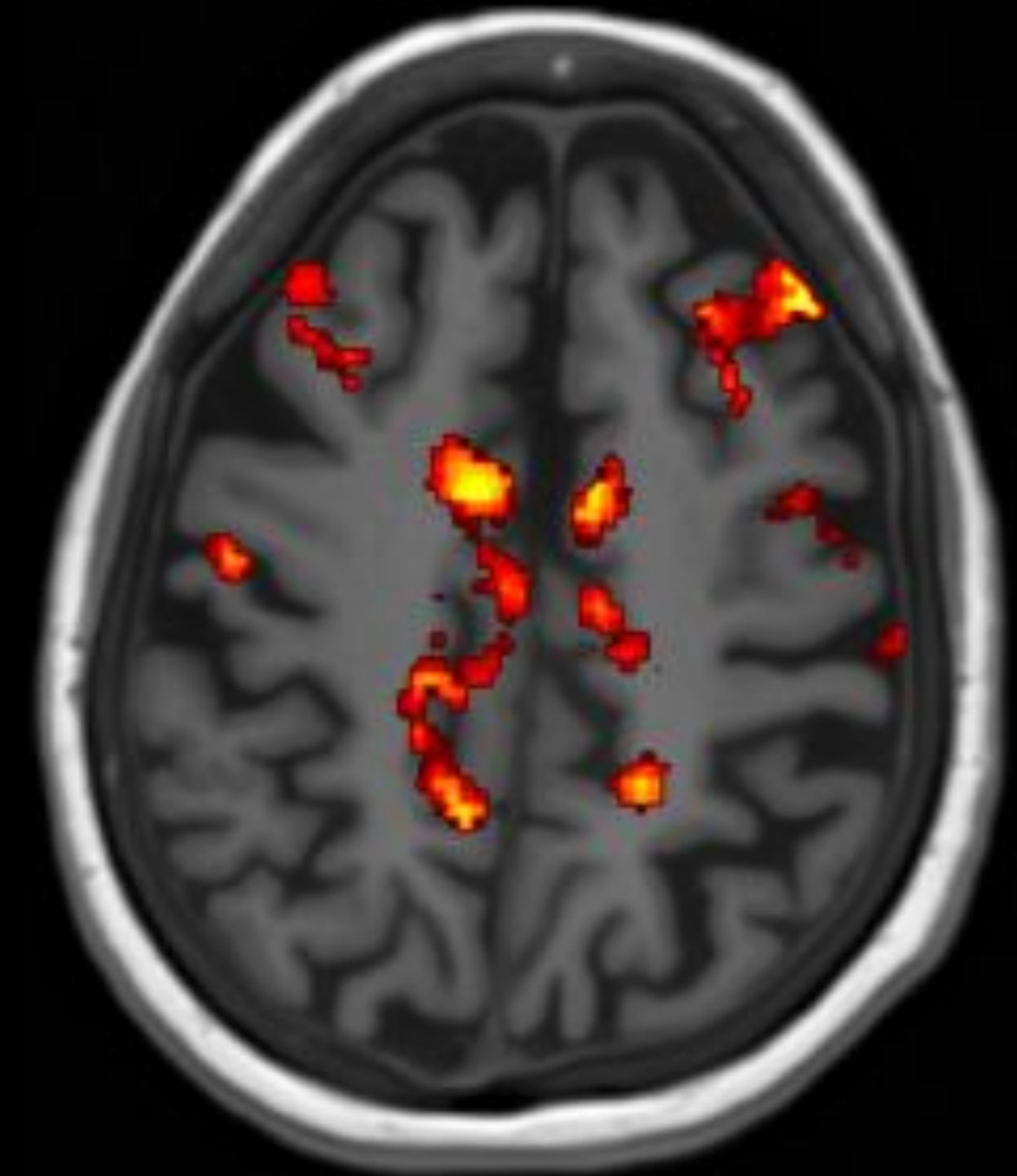
DAMAGE CAUSES

Reduced ability to prioritize stimuli and emotional dysregulation

KEY REGIONS

Anterior Insula

Anterior Cingulate Cortex (ACC)



KEY PATHWAYS

Frontal Aslant Tract (FAT)

Ventral Attention Network (VAN)

Cognitive Networks

FUNCTION

Detects unexpected stimuli and redirects attention

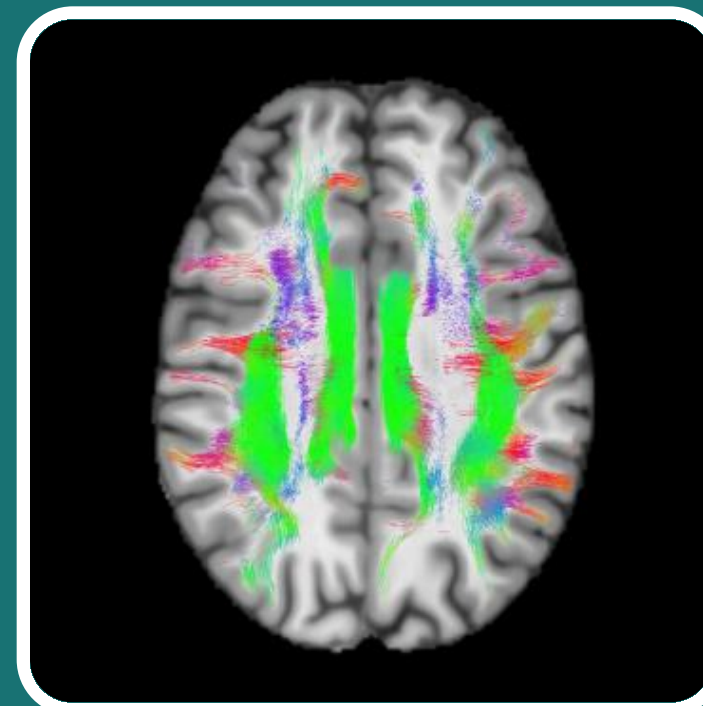
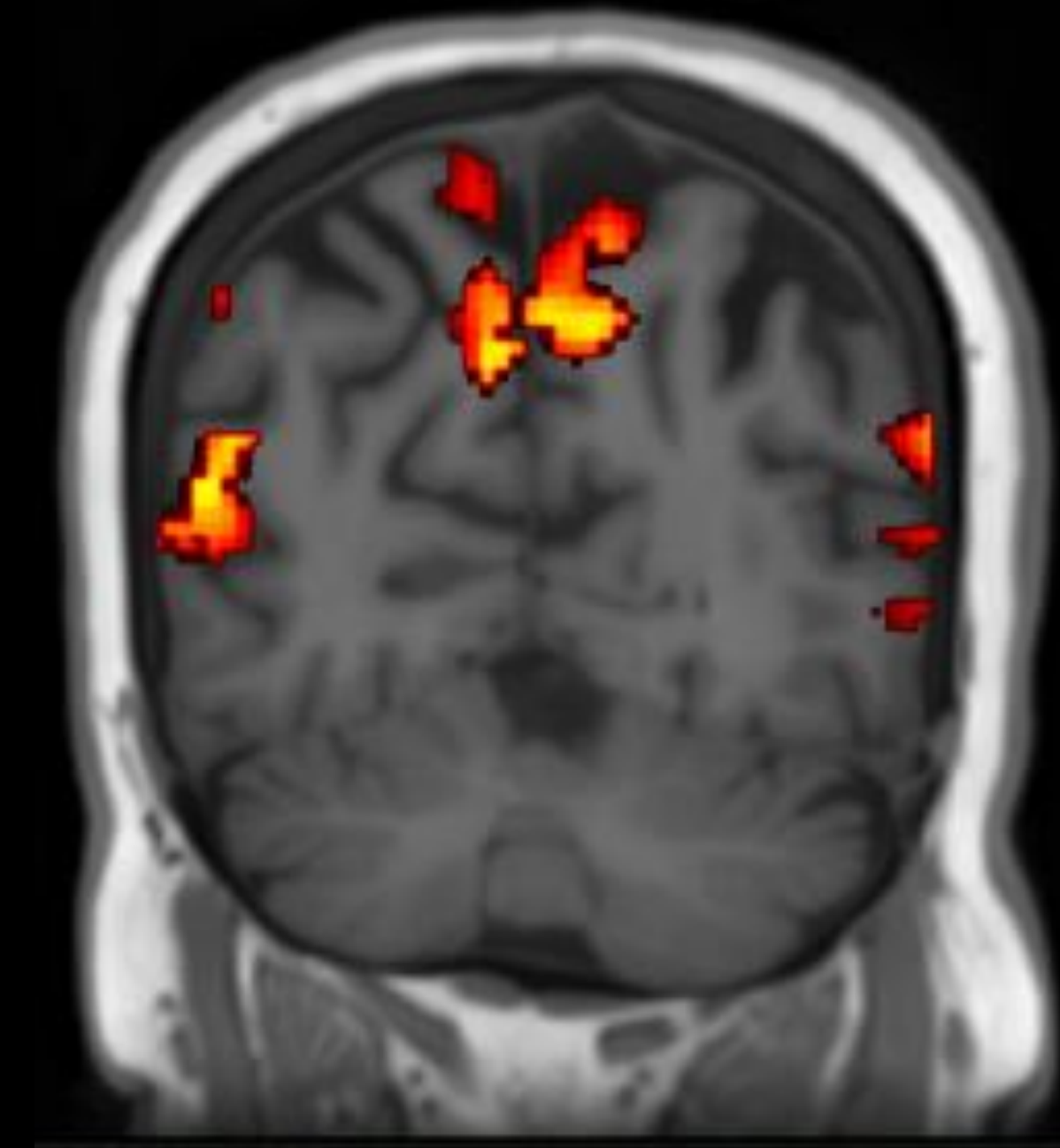
DAMAGE CAUSES

Deficits in shifting attention or increased distractibility

KEY REGIONS

Temporoparietal Junction (TPJ)

Ventrolateral Prefrontal Cortex



KEY PATHWAYS

Superior Longitudinal Fasciculus

Dorsal Attention Network (DAN)

Cognitive Networks

FUNCTION

Supports voluntary, sustained attention to tasks

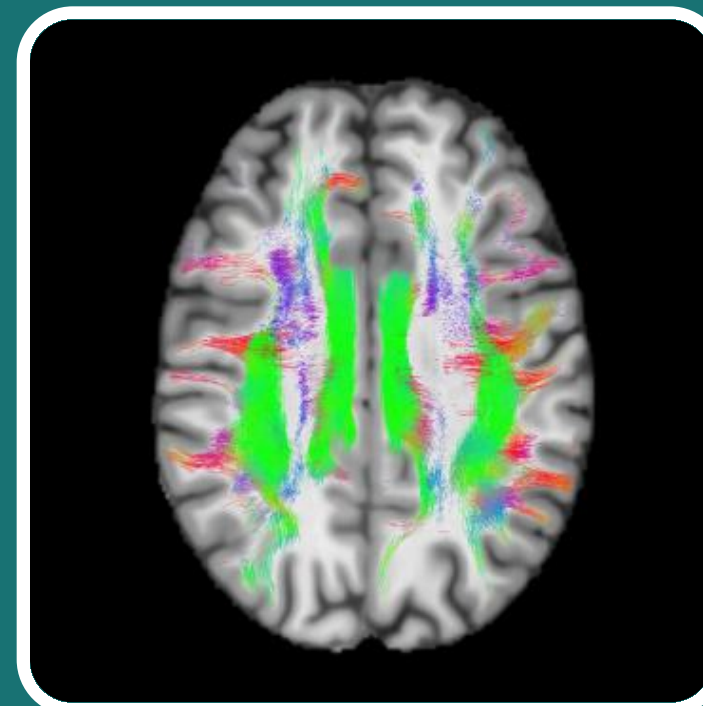
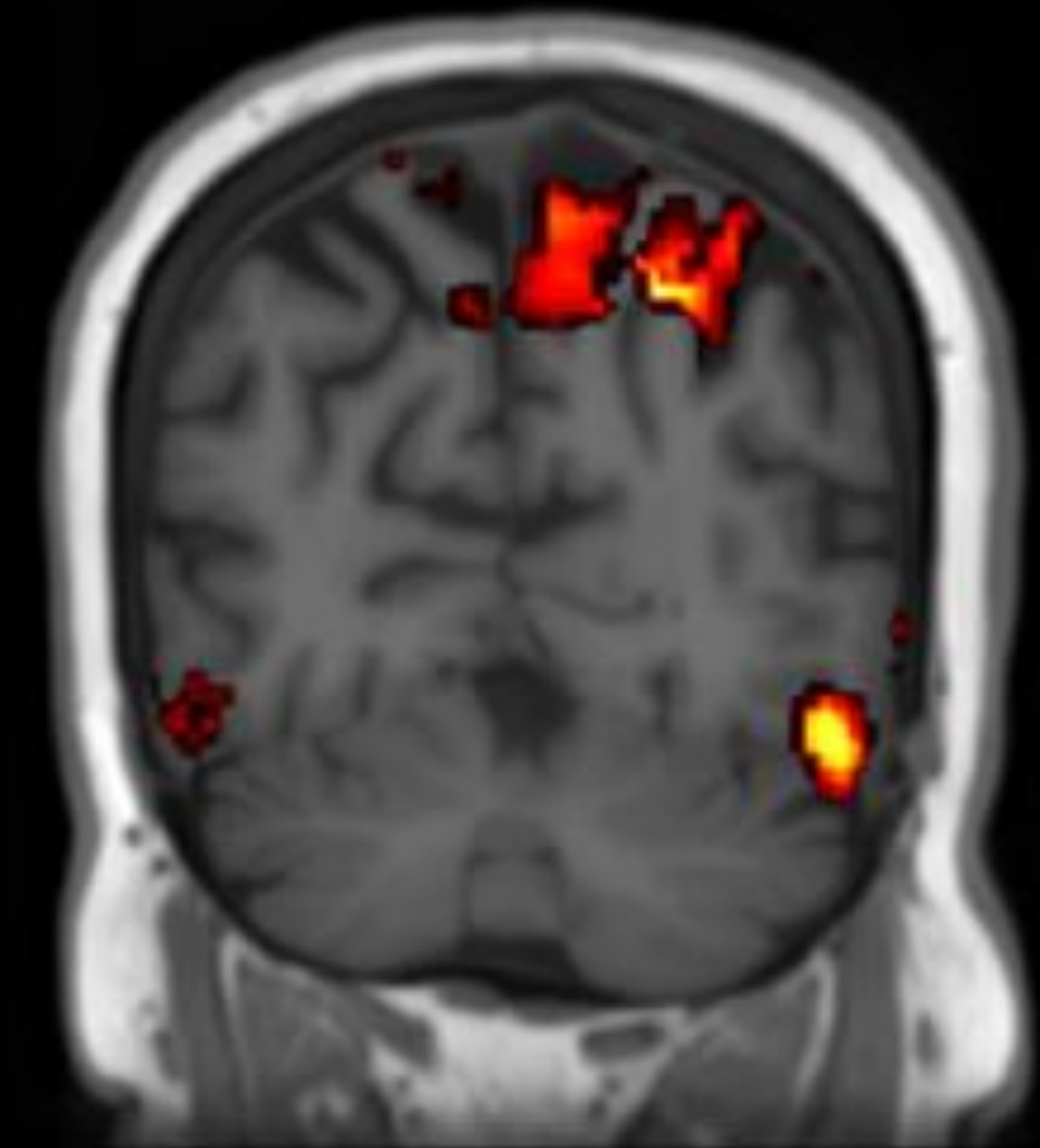
DAMAGE CAUSES

Difficulty in sustaining focus or poor task performance

KEY REGIONS

Frontal Eye Fields

Superior Parietal Lobule



KEY PATHWAYS

Superior Longitudinal Fasciculus

Thank you

Hope you found the toolkit helpful